

GraphRAG Makes it Possible to Digest Convolved Legal Jargon

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Abstract—Legal jargon easily overwhelms those who lack proper training. Aspiring entrepreneurs may feel intimidated by the complex web of regulations between them and their goals. Even when available, crucial information can be extensive and difficult to navigate without costly professional help. This article leverages recent advancements in the field of Natural Language Processing (NLP) to investigate the potential of Large Language Models (LLMs) in helping stakeholders understand the legal requirements of business aspirations. Inspired by [7], we compare 2 Machine Learning (ML) retrieval methods—Retrieval Augmented Generation (RAG), and Graph Retrieval Augmented Generation (GraphRAG)—assessing their performance on 2 sets of LLM generated queries. More specifically, our study evaluates 6 sub-models—4 from Light Retrieval Augmented Generation (LightRAG) and 2 from Microsoft’s Graph Retrieval Augmented Generation (MGRAG). We examine these sub-models across 4 key criteria—comprehensiveness, diversity, empowerment, and directness—as well as a combination of all 4 which we call “overall”. After 3 separate experiments, we observe that GraphRAGs have a clear advantage over RAGs for digesting complex language in this task. Within GraphRAGs, we observe that MGRAG has a slight advantage over LightRAG and naïve RAG. We present the results in grouped bar charts to offer clear and accessible comparisons, enabling anyone interested to quickly make informed decisions about which methods best suit their business needs.

Index Terms—Legal Compliance, Business Law, Real Estate, Zoning, Natural Language Processing (NLP), Large Language Models (LLMs), Graph Retrieval Augmented Generation, RAG, Entrepreneur

I. INTRODUCTION

The mountains of red tape that demand compliance can quickly douse the fire of entrepreneurial enthusiasm and stifle the desire to bring visions to life. Business/legal documentation is often coated in jargon that makes comprehension a fever dream for the untrained. Although reading legal documentation can be a lengthy and frustrating exercise, even for trained individuals, advancements in Artificial Intelligence (AI) and Machine Learning (ML) have empowered people to chase dreams they previously could not imagine. The field of Natural Language Processing (NLP), for example, a subset of AI/ML, is geared toward developing algorithms that can process, generate, or understand natural language. From advanced text to speech applications to smart assistants on phones and other devices, NLP enables a wide range of possibilities [4], [9]–[12].

In this era of rapidly expanding data lakes, the sheer volume of available information can overwhelm and paralyze newcomers before they know where to begin or what questions to ask. Making vast amounts of information available, alone, is not enough [7], especially when that information is convoluted. Even after investing considerable time and effort, stakeholders can still find themselves confused or uncertain. Furthermore, they may lack funding or resources to guide them through complex business and legal jargon. We aim to understand how Large Language Models (LLMs) can be applied to help in this area.

Researchers in [7] use disjoint research article titles and abstracts to illustrate the possibility of Retrieval Augmented Generation (RAG) and Graph Retrieval Augmented Generation (GraphRAG) pipelines replacing or enhancing Bibliographic Indexing/Databasing (BI/D). Our work builds on theirs, exploring the applicability of RAG pipelines in entrepreneurial and legal contexts. We focus particularly on real estate development. In conformity to current state of the art practice [1] [3], we apply LLM frameworks to analyze and answer 2 new sets of LLM-generated questions. Our application of this methodology focuses on a homogeneous collection of publicly available documents that outline the land and property regulations in Prince George’s County, Maryland [8]. While [7] was geared toward pure research, our work focuses on business/legal applications. Also, unlike their work, which is enabled by disjoint pieces of text, we use text from the same document (the Prince George’s County, MD Zoning Ordinance, Subdivision Regulations, & Landscape Manual (PGZO)).

II. RELATED WORKS

Conventional LLMs perpetually need to be trained or re-trained on new data to continue providing accurate information. This training process can be costly, time consuming, or just inconvenient. The RAG framework/pipeline allows LLMs to answer accurately without needing to be retrained. Its implementation, though somewhat technical, is fairly straightforward.

Users first provide source documents to the pipeline. It splits the documents into chunks and embeds the chunks in a certain way [6]. The embeddings and chunks are stored in a data

User Query:
What are the top 5 themes in the data?

RAG Response:
Based on the analysis of the dataset, the following are the top five themes identified: ### 1. **Integration of AI Technologies** The integration of artificial intelligence (AI) across various sectors, including healthcare, education, and legal advisory services, is a prominent theme. This integration highlights the transformative role of AI in enhancing operational efficiency, decision-making, and user interactions. Notably, frameworks like Retrieval-Augmented Generation (RAG) and models such as ChatGPT are emphasized for their applications in improving service delivery and accuracy in various fields [Data: Reports (2496, 2017, 2827, 3326, 662)]. ### 2. **Advancements in Large Language Models (LLMs)** The development and optimization of large language models (LLMs) are central to many discussions. These models are being integrated into applications ranging from healthcare to customer service, showcasing their versatility and significance in enhancing communication and information retrieval capabilities. The focus on frameworks like RAG and methodologies for fine-tuning LLMs indicates a strong interest in improving their performance and reliability [Data: Reports (2014, 2529, 2140, 2217, 3504)]...

Fig. 1. Snippet of a RAG model's response.

container (not the LLM). When a query is submitted, it is first embedded using the same embedding method. A retrieval engine then compares the query embeddings to data chunk embeddings in order to identify the most relevant data chunks. These chunks are combined with the original query to create a context package, which is finally sent to the LLM. With this context package—the query and its matching data chunks—the LLM can generate a more informed/relevant response [5]. Figure 1 displays sample output from a RAG pipeline.

There are countless variations which aim to improve traditional RAG pipelines. Retrieval Augmented FineTuning (RAFT), for example, allows RAGs to ignore “distractor”

documents, leading to more efficient generation in some cases [14]. The Hypothetical Document Embeddings (HyDE) method transforms user queries into LLM generated documents before embedding and using those generated documents for vector search instead of the original query [2]. Despite these advancements, however, RAG and other models that rely purely on chunk embedding comparison (also known as vector search) are said to be limited [3]. Some queries require a more holistic understanding of the dataset [1].

GraphRAG pipelines break document chunks even further down to extract entities and relationships. A sentence like “the fat cat jumped over the lazy dog”, which would all be stored as a singular entity in traditional RAGs, could be broken into entities and their corresponding relationship information (e.g. cat - fat, dog - lazy, cat - jumps over - dog, etc.). GraphRAGs have an enhanced ability to analyze the overall structure and content of data since its representation is no longer flat (chunk-focused). “When handling large token counts and complex queries that require a thorough understanding of the dataset’s context, graph-based systems... consistently outperform purely chunk-based retrieval methods such as [RAG] and HyDE” [3]. Example queries GraphRAGs, as opposed to RAGs, would be more suited to answer can be found in Figure 4.

Our attention now shifts to two GraphRAG implementations: Microsoft’s Graph Retrieval Augmented Generation (MGRAG) and Light Retrieval Augmented Generation (LightRAG). While MGRAG augments the retrieval process by including summaries of entities and relationships into the final assessment, LightRAG focuses on reducing the retrieval overhead from the community-based traversal method used in MGRAG [3].

III. PROPOSED APPROACH

The following experiments draw inspiration from the work of [7]. Whereas they use disjoint article abstracts as the backbone of their research, the data used in our experiments are homogeneous (all from PGZO). Our methods include slight changes to fit the specific zoning/business/legal use case.

Our main objectives include:

- 1) To apply RAG and GraphRAG to answer zoning/housing/real estate-specific queries, using only information found within the PGZO.
- 2) To compare RAG and GraphRAG sub-model performance on five criteria (comprehensiveness, diversity, empowerment, directness, overall).
- 3) To identify best RAG or GraphRAG sub-model for answering queries when context is provided.
- 4) To identify the best RAG or GraphRAG sub-model for answering queries when context is not provided.
- 5) To evaluate RAG and GraphRAG sub-model performance on queries that require semantical understanding of the entire data pool.

A. Experiment 1

According to the framework used in [7], all experiments follow the same general flow (illustrated in Figure 3) with

minute changes to address specific objectives. In experiment 1, we simultaneously send all 6 sub-model answers for a question into an arbitrary gpt-4o-mini instance in the following order: `l_z_h`, `m_z_g`, `m_z_l`, `l_z_g`, `l_z_n`, `l_z_l`. The first character of each name in this nomenclature indicates LightRAG or MGRAG and the last character indicates the type of sub-model: naïve, local, global, or hybrid. The LLM's instructions are to rank answers from best to worst, making sure it provides reasoning for each ranking in the provided JSON format.

B. Experiment 2

Experiment 2 introduces a randomized tournament bracket to diminish bias from experiment 1 (see Figure 2 for tournament details). After all answers for a specific query are generated, before being sent to the judge LLM, their order is randomized to obtain an initial seeding. A full run through the bracket marks the judging of a singular question's 6 answers. Bracket steps are repeated for all questions, with and without context. A loser's bracket is included to ensure that all query responses have a chance to be tested for each attribute.

We also remove language that could add bias from the judge LLM's prompt. Rather than referring to the answers as "Answer 1" or "Answer 2", for example, we refer to them by the string representation of the answers origin from experiment 1—`l_z_h`, `m_z_g`, `m_z_l`, `l_z_g`, `l_z_n`, or `l_z_l`. With updated instructions and output format, experiment 2 generates more detailed data files. Each file contains the comparison of two answers to the same question evaluated on a specific criterion rather than comparing all six answers as in experiment 1. This method also produces more comparisons per context pool—9 rounds per question $\times$ 125 questions $\times$ 1 file per round = 1136 files comparisons instead of the 126 comparisons of experiment 1.

C. Experiment 3

Tasks like "What are the top 5 themes in the dataset?" are difficult for chunk focused retrievers because they search for document chunks whose vector embeddings closely match the query "What are the top 5 themes in the dataset?" rather than searching for chunks that respond to that phrase as a question. Experiment 3 questions were hand-designed to be difficult to enhance with RAG customizations. We repeat the same steps from the other experiments to fairly test all six sub-models and rank their responses. Samples of the hand-crafted questions are displayed in Figure 4.

We repeat the same steps found in the other experiments to test all 6 sub-models and rank their responses.

D. Data Collection

We downloaded all documents from the Prince George's County, MD Zoning Ordinance, Subdivision Regulations, & Landscape Manual (PGZO) [8].

E. Question Generation

In alignment with contemporary best practice for this approach, we give an arbitrary LLM, a gpt-4o-mini instance, a

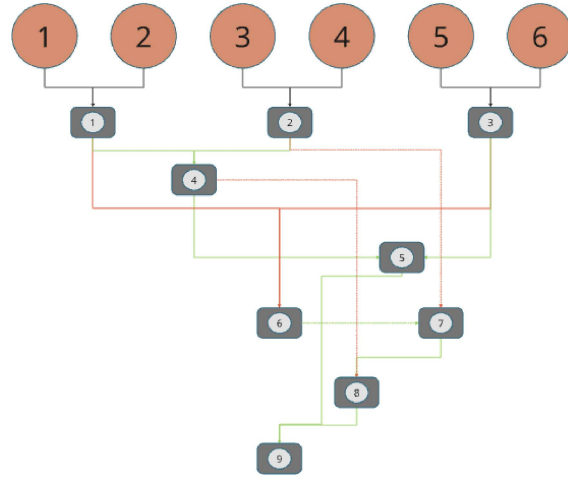


Fig. 2. This figure shows the bracket used to test 3 pairs of answers per query. Each answer (from a sub-model) has a randomized seed, 1, 2, 3, 4, 5, or 6. Seeds 1 and 2 face each other in round 1, followed by seeds 3 and 4 in round 2, and seeds 5 and 6 in round 3. The winner of round 1 moves to round 4 where it faces the winner of round 2. The winners from round 4 and 3 face each other in round 5. Round 6 starts the losers bracket, having the loser of round 1 face the loser of round 3. The winner of this matchup moves to round 7 where it faces the loser of round 2. Round 8 features a matchup between the winner of the losers bracket and the loser of round 4. The winner of this round faces the winner of the winners bracket in the final round, 9. The color red denotes a loss while green highlights a win.

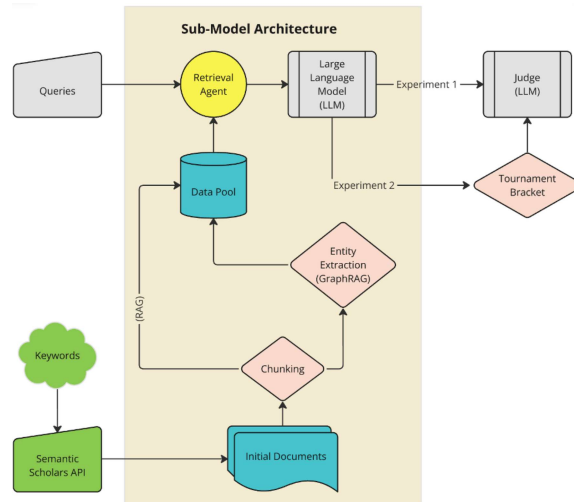


Fig. 3. This figure shows a flowchart of the experiments contained within this paper. Experiment 3 is not explicitly displayed because though it goes through the tournament bracket, the real difference is at the creation of queries. Visit subsection Experiment 3 for more information.

1. Provide an overview of the entire dataset.
2. What are the key patterns or trends in the data over time?
3. What are the 5 most mentioned topics in this dataset. How many times is each mentioned and where?
4. How do different categories or groups in the data compare against each other?
5. Can you summarize the key insights across different subgroups in the data?

Fig. 4. Sample of experiment 3 questions.

description of the data and have it generate 125 questions [3] [1] [7]. We adapt this set into 2 separate pools, each containing 125 queries. Set 1 contains a user, task, and question, forming our “with context” pool, while Set 2 contains only the question, serving as our “without context” pool. For example, if a query from set 1 is “User is a Planning Director whose task is to analyze the impact of proposed subdivisions on school enrollment. Their question is how do historical data trends correlate with the projected enrollment figures based on new subdivision applications?”, then the complementary set 2 query only contains the latter part, namely, “How do historical data trends correlate with the projected enrollment figures based on new subdivision applications?”. Once queries are generated, we pass them into 6 different sub-models—4 LightRAG sub-models and 2 MGRAG sub-models—and catalog their answers.

F. Hypothesis

We hypothesize that hybrid LightRAG will perform the best in the “overall” criteria due to its innovative implementation of high and low level information retrieval. We believe the ability to synthesize high and low level information will not only be important for responding to queries with and without context, but will also aid in answering questions that require deep semantical comprehension.

IV. PERFORMANCE EVALUATION

A. Experiment 1

Figure 5 displays the results of experiment 1, where the judge LLM is given all six sub-model responses in the same prompt and asked to rank them by best response according to a criterion. For comprehensiveness, hybrid LightRAG overwhelmingly dominated the other models, winning 126 comparisons with context and 121 comparisons without context. Results were slightly more varied for diversity. The hybrid LightRAG model won the most comparisons—84 with context and 61 without context. hybrid LightRAG also drastically outshone the rest in empowerment, winning 118 comparisons

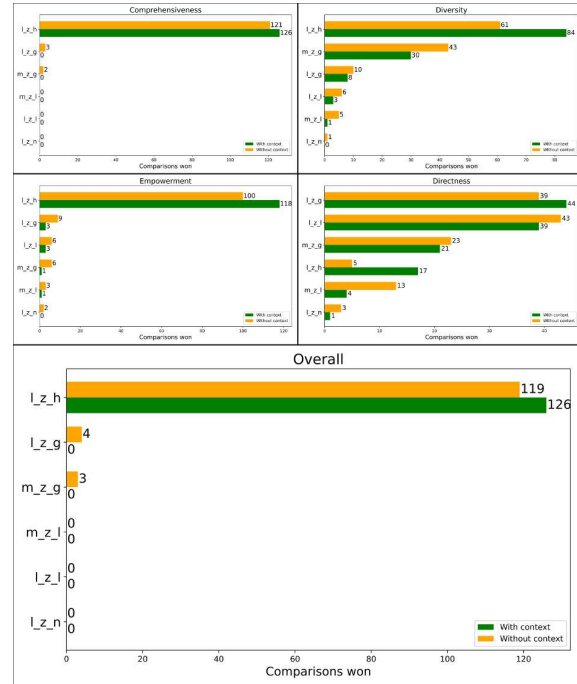


Fig. 5. This Figure shows experiment 1 Results. The judge LLM is given all 6 sub-model responses in the same prompt and asked to rank them by best response according to a criterion. The highest ranking model for each query is cataloged. Each bar in this chart represents the aggregate number of wins that model has per criterion on LLM-generated queries with and without context. The comprehensiveness criteria graph is at the top left, diversity at the top right, empowerment in the middle left, directness in the middle right, and overall at the bottom. The “l_z_h” model, aka hybrid LightRAG, is clearly dominant across most criteria. Other category placements are discussed in experiment 1 Results subsection.

with context and 100 comparisons without context. Global LightRAG won the directness criteria, winning 44 comparisons with context and 39 without context. It was closely followed by local LightRAG which won 39 comparisons with context and 43 comparisons without context. Hybrid LightRAG completely eclipsed the competition for the overall criteria, winning 126 comparisons with context and 119 comparisons without context.

B. Experiment 2

Results for experiment 2 can be found in Figure 6. While each bar is still an aggregate of that model’s wins per criteria, the judge LLM only receives two sub-model responses at a time for comparison. This experiment makes use of the randomized seeds and tournament bracket (see Figure 2). For comprehensiveness, hybrid LightRAG won 256 comparisons with context and 243 comparisons without context. Global MGRAG was a close second, winning 245 comparisons with

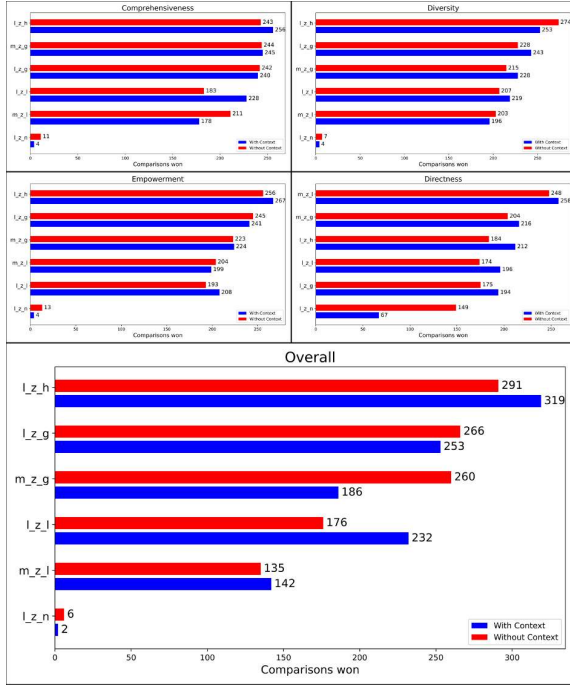


Fig. 6. This figure shows experiment 2 Results. The judge LLM is given 2 sub-model responses at a time and asked to rank them by best response according to a criterion. The higher ranking response's model is cataloged. Each bar in this chart represents the aggregate number of wins a model has per criterion on LLM-generated queries with and without context. The comprehensiveness criteria graph is at the top left, diversity at the top right, empowerment in the middle left, directness in the middle right, and overall at the bottom. Winners of each category are discussed in experiment 2 Results subsection.

context and 244 comparisons without context. For diversity, hybrid LightRAG won 253 rounds with context and 274 rounds without context. Global LightRAG came in second, winning 243 rounds with context and 228 rounds without context. In terms of empowerment, hybrid LightRAG won 267 rounds with context and 256 rounds without context. It was followed by global LightRAG which won 241 rounds with context and 245 rounds without context. Local MGRAG won the directness criteria, winning 258 rounds with context and 248 rounds without context. It was followed by global MGRAG which won 216 rounds with context and 204 rounds without context. Hybrid LightRAG again had great performance on the overall criteria, winning 319 rounds with context and 291 rounds without context. It was followed by global LightRAG which won 253 rounds with context and 266 rounds without context.

C. Experiment 3

Figure 7 displays the results of experiment 3, which also uses randomized seeds and the tournament bracket (see Fig-

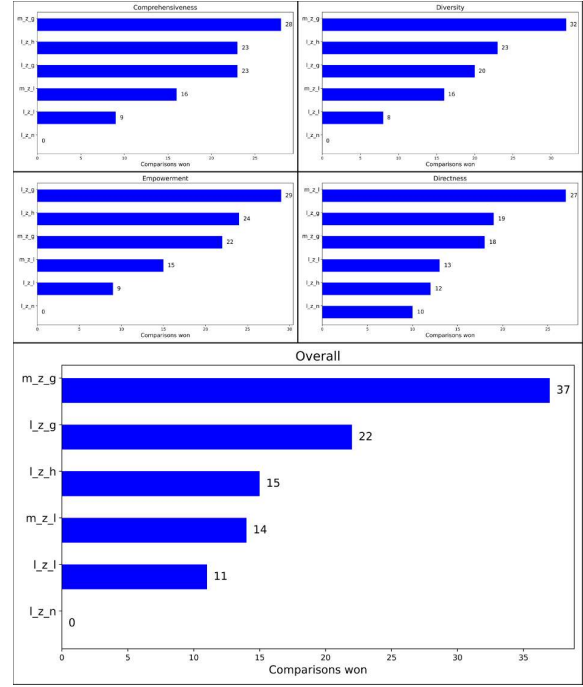


Fig. 7. This figure shows experiment 3 Results. The judge LLM is given 2 sub-model responses at a time and asked to rank them by best response according to a criterion. The higher ranking response's model is cataloged. Each bar in this chart represents the aggregate number of wins a model has per criterion on hand-generated queries without context. Comprehensiveness is at the top left, diversity at the top right, empowerment in the middle left, directness in the middle right, and overall at the bottom. Winners of each category are discussed in experiment 3 Results subsection.

ure 2) on the hand-generated queries. Global MGRAG won the comprehensiveness category, coming out on top in 28 comparisons. It also dominated the diversity criteria, winning 32 rounds. empowerment wins were led by global LightRAG, which won 29 comparisons. Local MGRAG won the directness criteria with 27 comparison wins. In the final overall criteria assessment, global MGRAG had 37 comparison wins, followed by global LightRAG which won 22 comparisons.

V. DISCUSSION

Results from experiment 1 would normally indicate that hybrid LightRAG is the dominant model. However, we found it odd that seldom did the other GraphRAG variations win comparisons. Upon further investigation, we were led to believe that this is model skew toward the first answer given to the judge LLM. Contemporary research supports our assessment of this being bias [13]. We revised experiment 1 into experiment 2 in hopes of reducing that bias. Results were immediately more evenly distributed, suggesting that our mitigations against bias were effective (Figure 6). As expected,

though it still won some comparisons, naïve RAG performed poorly in this experiment.

Since they were generated by an LLM, it is conceivable that questions from the experiment 2 test pool were trivially answerable and did not require in depth semantical understanding of the dataset [7]. We conducted experiment 3 to address this concern. Naïve RAG won no comparisons in this case, indicating that GraphRAGs are better for answering complex queries which require semantical understanding of large data pools.

Our hypothesis proved only partially true. Hybrid Lightrag, which dominated experiments 1 and 2 criteria comparisons, was overtaken by global MGRAG when the questions required an in depth understanding of the data pool—a.k.a. experiment 3. This was arguably the most important experiment for this specific use case. MGRAG seems to perform better. Though our hypothesis was only partially true, we were still able to address the initial goal of our research—to explore the applicability of LLMs in this space.

Using a similar framework to [7] on the PGZO [8] we were able to get answers to zoning/housing/real estate specific questions without needing to train or re-train an LLM; thereby addressing objective 1. Our graphs address objectives 2, 3, and 4, displaying the ranked results of RAG and GraphRAG sub-model performance on all 5 criteria with and without context. Finally, experiment 3 demonstrates GraphRAG’s ability to tackle subject matter-specific questions that demand deep semantic understanding of data (objective 5).

A. Future Work

These experiments highlight the strengths of RAG pipelines and show the potential of RAG extensions like GraphRAG and HyDE. Similar pipelines along with their complementary extensions and frameworks can be leveraged for other purposes. It is conceivable that they be used to track the timeline and development of ideas in a data pool. It would allow users to see where and when an idea originated as well as how it was updated and modified over time. Additionally, since they have an enhanced ability to understand and perform complex logical reasoning, and [7] showed their applicability on disjoint data, it seems feasible to use GraphRAGs for live fact checking. Such a pipeline would take preliminary data into its knowledge base and check all incoming queries against it.

VI. CONCLUSION

While RAG models introduce the ability to answer difficult document-based questions, even those that typically require subject matter expertise, they struggle when the questions necessitate elevated levels of logical interpellation. GraphRAG improves on RAG by using entity and relationship extraction instead of general text chunking and vector search, allowing it to understand and respond to questions that cannot be edited to become more RAG retriever-friendly through customizations like HyDE.

Volumes of pertinent legal and business documentation are publicly available, but entrepreneurs still struggle to navigate

the convoluted language in which they are presented. Additionally, sifting through the data can be very time consuming and ultimately unproductive without proper training. Getting personalized professional help may require resources that are not readily available to the budding entrepreneur. AI/ML advancements in the field of NLP simply open the doors of possibility. Of the ones we tested, the MGRAG pipeline outperformed the rest in the most relevant category.

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